

ISA 419: Data-Driven Security

16: Machine Learning Regression Models

Fadel M. Megahed, PhD

Professor
Farmer School of Business
Miami University

 @FadelMegahed

 fmegahed

 fmegahed@miamioh.edu

 Automated Scheduler for Office Hours

Spring 2025

Quick Refresher of Last Class

- ✓ Describe the basic concepts of regression analysis, including the roles of independent and dependent variables.
- ✓ Assess regression models using metrics like R-squared and Mean Squared Error and interpret the results.
- ✓ Describe the two modeling mindsets: explanatory and predictive.

Learning Objectives for Today's Class

- Recognize the differences and similarities between traditional regression and machine learning regression models
- Describe advanced techniques like Ridge and Lasso
- Explore non-linear models like Decision Trees and Random Forests for regression tasks
- Apply machine learning regression models to cyber security data sets

Differences Between Traditional (Statistical) and Algorithmic (Machine Learning) Modeling Cultures

Statistical Modeling Culture

Find a stochastic model of the data-generating process in the form of: $Y = f(X, parameters, \epsilon)$



Assumptions

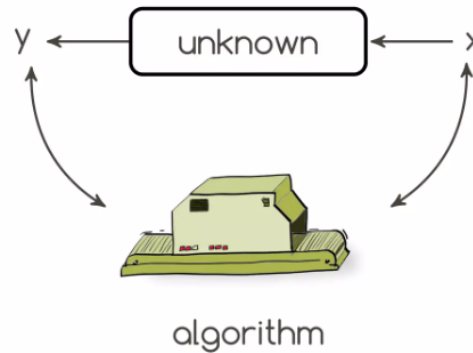
- Specific stochastic model of the data-generating process
- Distributional assumptions about the errors
- Linearities
- Manual specification of interactions

Problems

- Conclusions about model, and not about "nature"
- Assumptions often violated
- Often no model evaluation (no focus on prediction)

Algorithmic (Machine Learning) Modeling Culture

Find a function $f(x)$ that minimizes the loss $L(Y, f(X))$



Assumptions

- No assumptions about the data-generating process (true mechanism of generating the data is **unknown** and not of direct interest).
- No assumptions about the distribution of the errors.

Problems

- Some models are "black boxes".
- Stability of the model is not guaranteed (and honestly this is also a problem in statistical modeling).

Statistical Modeling: The Two Cultures

Statistical Science
2001, Vol. 16, No. 3, 199–231

Statistical Modeling: The Two Cultures

Leo Breiman

Abstract. There are two cultures in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown. The statistical community has been committed to the almost exclusive use of data models. This commitment has led to irrelevant theory, questionable conclusions, and has kept statisticians from working on a large range of interesting current problems. Algorithmic modeling, both in theory and practice, has developed rapidly in fields outside statistics. It can be used both on large complex data sets and as a more accurate and informative alternative to data modeling on smaller data sets. If our goal as a field is to use data to solve problems, then we need to move away from exclusive dependence on data models and adopt a more diverse set of tools.

1. INTRODUCTION

Statistics starts with data. Think of the data as being generated by a black box in which a vector of input variables $\mathbf{x}$ (independent variables) go in one side, and on the other side the response variables $\mathbf{y}$ come out. Inside the black box, nature functions to associate the predictor variables with the response variables, so the picture is like this:



There are two goals in analyzing the data:

Prediction. To be able to predict what the responses are going to be to future input variables;

Information. To extract some information about how nature is associating the response variables to the input variables.

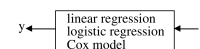
There are two different approaches toward these goals:

The Data Modeling Culture

The analysis in this culture starts with assuming a stochastic data model for the inside of the black box. For example, a common data model is that data are generated by independent draws from
response variables = $f(\text{predictor variables, random noise, parameters})$

Leo Breiman is Professor, Department of Statistics, University of California, Berkeley, California 94720-4735 (e-mail: leo@stat.berkeley.edu).

The values of the parameters are estimated from the data and the model then used for information and/or prediction. Thus the black box is filled in like this:

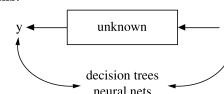


Model validation. Yes–no using goodness-of-fit tests and residual examination.

Estimated culture population. 98% of all statisticians.

The Algorithmic Modeling Culture

The analysis in this culture considers the inside of the box complex and unknown. Their approach is to find a function $f(\mathbf{x})$ —an algorithm that operates on $\mathbf{x}$ to predict the responses $\mathbf{y}$. Their black box looks like this:



Model validation. Measured by predictive accuracy.

Estimated culture population. 2% of statisticians, many in other fields.

In this paper I will argue that the focus in the statistical community on data models has:

- Led to irrelevant theory and questionable scientific conclusions;

Advanced Techniques for Regression

Ridge Regression

RIDGE REGRESSION

Residual sum of
squares $\rightarrow$

RSS

$$+ \lambda \sum_{j=1}^p \hat{\beta}_j^2$$

Parameters squared

Tuning parameter

Shrinkage

Remember:
Standardize
the data first.

Chris Albon

Disadvantage:
parameters cannot
be zero like
with Lasso
regression.

Lasso Regression

LASSO

FOR FEATURE SELECTION

- Lasso regression uses L1 norm as regularizer.

$$\alpha \sum_{i=1}^K |w_i|$$

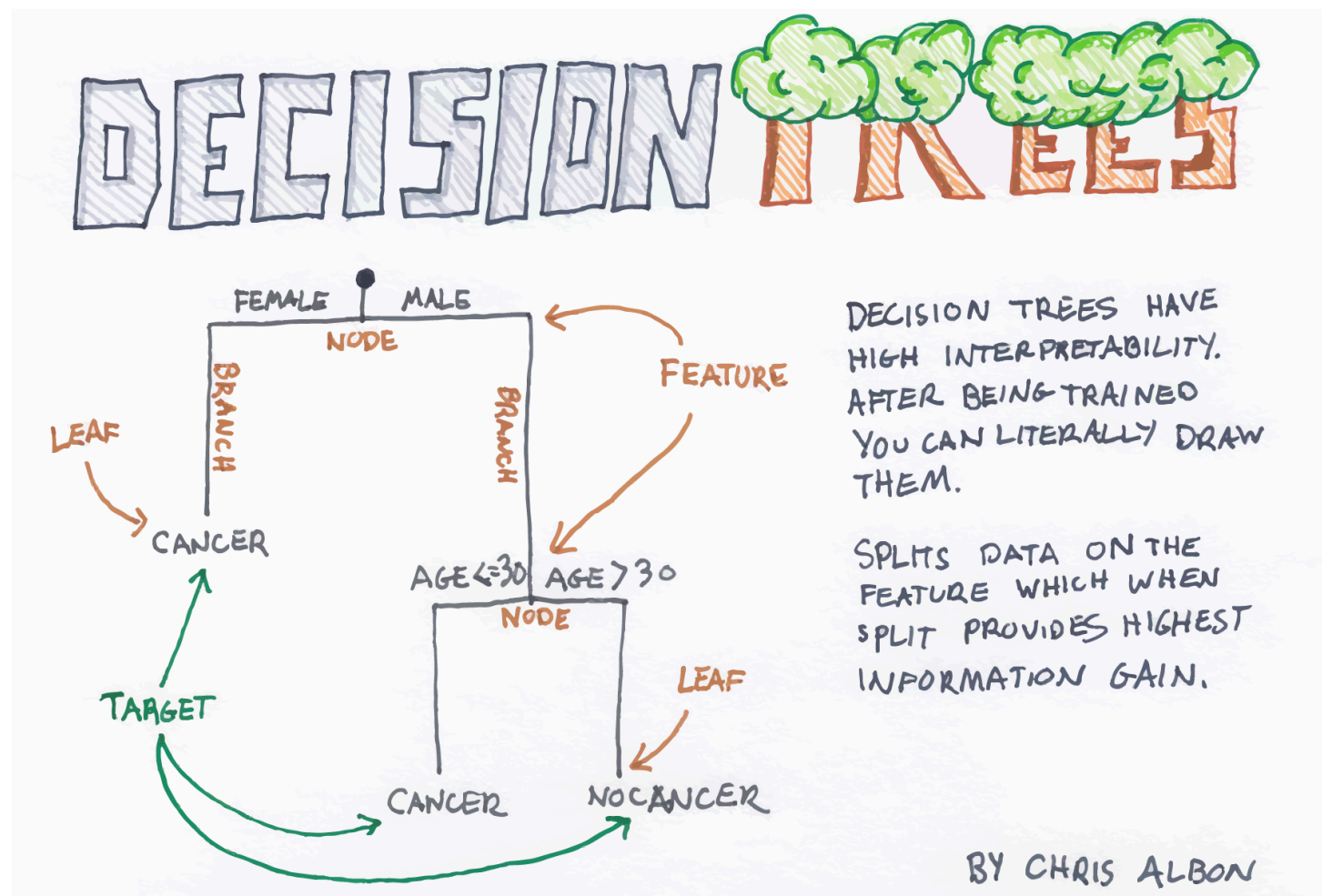
alpha $\uparrow$ w_i $\uparrow$ parameter

- Unlike ridge regression, lasso's norm regularizer drives parameters to zero.
- Higher the value of alpha, the fewer features have non-zero values.

Chris Albon

Non-Linear Models for Regression

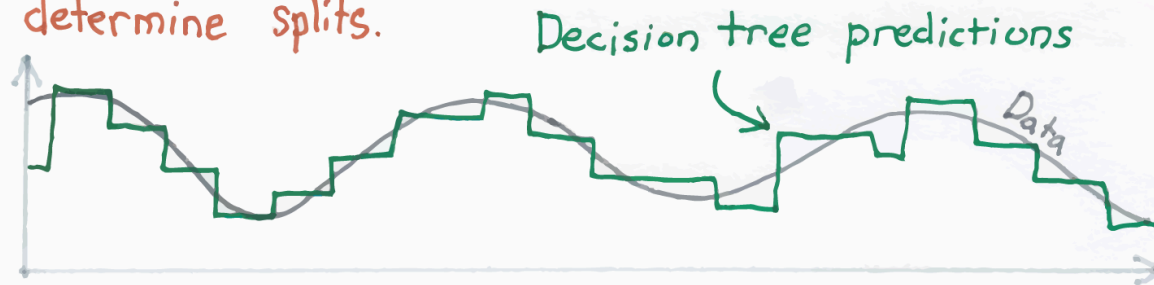
Decision Trees (Classification and Regression Trees)



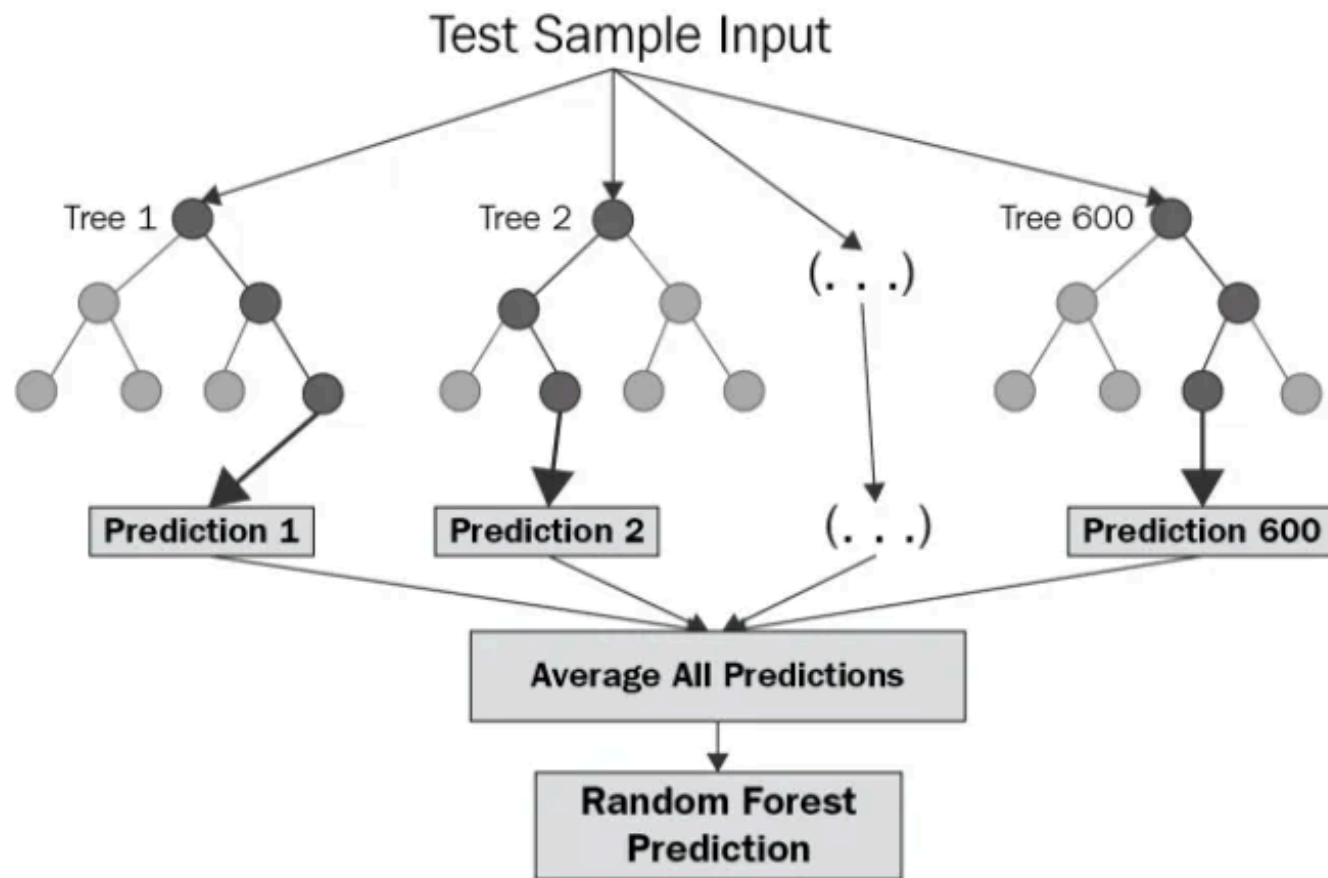
Decision Trees (Classification and Regression Trees)

DECISION TREE REGRESSION

Similar to decision tree classification, however uses Mean Squared Error or similar metrics instead of cross-entropy or Gini impurity to determine splits.



Random Forests



Random Forests

THE RANDOM IN RANDOM FOREST

1. Each tree gets a random sample of observations with replacement.
2. Each tree gets all features, but at each node only a subset of those features are available.

Chris Albon

Applications to Cybersecurity Datasets

Example: Predicting Flow Duration in IoT Networks

```
from ucimlrepo import fetch_ucirepo

# fetch dataset
rt_iot2022 = fetch_ucirepo(id=942)

# data (as pandas dataframes)
X = rt_iot2022.data.features
y = rt_iot2022.data.targets
df = pd.concat([X, y], axis=1)

# subset the features and dataset
df = (
    df
    .sample(frac=0.1, random_state=2025)
    .reset_index(drop=True)
    .loc[:, ['proto', 'service', 'fwd_pkts_payload.avg', 'bwd_pkts_payload.avg', 'fwd_subflow_
```

Recap

Summary of Main Points

By now, you should be able to do the following:

- Recognize the differences and similarities between traditional regression and machine learning regression models
- Describe advanced techniques like Ridge and Lasso
- Explore non-linear models like Decision Trees and Random Forests for regression tasks
- Apply machine learning regression models to cyber security data sets



Review and Clarification



- **Class Notes:** Take some time to revisit your class notes for key insights and concepts.
- **Zoom Recording:** The recording of today's class will be made available on Canvas approximately 3-4 hours after the end of class.
- **Questions:** Please don't hesitate to ask for clarification on any topics discussed in class. It's crucial not to let questions accumulate.